

Two-Way Analysis of Variance

STAT 24510-30040

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The ANOVA we have learned directly compares the means of several treatments, without considering the treatment structure. When the treatments consist of two or more treatment factors, two-way and multi-way ANOVA are applied.

Consider a completely randomized experiment with two treatment factors: factor A and factor B. Each treatment factor may have several levels, say, I levels for factor A and J levels for factor B. Instead of analyzing the data as one treatment with $g = I \times J$ levels, it is more efficient to take the factorial structure into consideration by using two-way layout analysis.

Two-way layout data

Outcome measurements under A level i and B level j are denoted as $y_{ij1}, y_{ij2}, \dots, y_{ijk_{ij}}$.

Two-way layout data can be displayed in an array. If factor A has levels $1, 2, \dots, a$, factor B has levels $1, 2, \dots, b$, and there are k_{ij} replicates for A level i and B level j , the data structure has the following form.

	B-level 1	B-level 2		B-level b
A-level 1	y_{111} $\vdots$ $y_{11k_{11}}$	y_{121} $\vdots$ $y_{12k_{12}}$	$\dots\dots$	y_{1b1} $\vdots$ $y_{1bk_{1b}}$
A-level 2	y_{211} $\vdots$ $y_{21k_{21}}$	y_{221} $\vdots$ $y_{22k_{22}}$	$\dots\dots$	y_{2b1} $\vdots$ $y_{2bk_{2b}}$
$\vdots$	$\vdots$	$\vdots$	$\dots\dots$	$\vdots$
A-level a	y_{a11} $\vdots$ $y_{a1k_{a1}}$	y_{a21} $\vdots$ $y_{a2k_{a2}}$	$\dots\dots$	y_{ab1} $\vdots$ $y_{abk_{ab}}$

Denote random variable Y_{ijk} as the k th observation with A factor level i and B factor level j .

A linear model for an experiment with two factors is

$$Y_{ijk} = \mu_{ij} + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \sim i.i.d. N(0, \sigma^2)$$

where

$i = 1, \dots, I$ Rows: levels of Factor or A
 $j = 1, \dots, J$ Columns: levels of Factor or B
 $k = 1, \dots, K_{ij}$ Replications at (i, j)

Other common notations:

Number of levels of factor A uses a instead of I ,

Number of levels of factor B uses b instead of J ,

Number of replicates within each treatment combination is r_{ij} or n_{ij} instead of K_{ij} .

Two-way layout parametrization

By the model assumption for $Y_{ijk} = \mu_{ij} + \varepsilon_{ijk}$, the cell mean for A-level i and B-level j is

$$\mathbb{E}(Y_{ij}) = \mu_{ij}$$

There are $I \times J$ many such μ_{ij} 's.

To reflect the two-factor design, μ_{ij} has intrinsic structures and is parametrized accordingly.

A linear model containing the main effect terms and the interaction is commonly parametrized with

$$\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$$

where

μ	Grand mean
α_i	Row effect (Factor A)
β_j	Column effect (Factor B)
$\gamma_{ij} = (\alpha\beta)_{ij}$	Interaction of A and B

Interaction of factors characterizes departure from additive effects of the factors.

The linear model $Y_{ijk} = \mu_{ij} + \varepsilon_{ijk}$ is then re-parametrized as

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \sim i.i.d. N(0, \sigma^2)$$

with $i = 1, \dots, I$, $j = 1, \dots, J$, $k = 1, \dots, K_{ij}$.

Remarks (on parameters of the two-way ANOVA model)

- The treatment effects are differential effects.
- Constraints: There are at most IJ (coefficient) parameters. Constraints are needed.

A common constraint is the zero-sum constraints:

$$\sum_{i=1}^I \alpha_i = 0, \quad \sum_{j=1}^J \beta_j = 0, \quad \sum_{i=1}^I \gamma_{ij} = 0, \quad \sum_{j=1}^J \gamma_{ij} = 0 \quad (1)$$

- Estimators

For balanced design with replicates $K_{ij} \equiv K$, under the sum-zero constraints, natural estimators are

$$\hat{\mu} = \bar{Y}_{\bullet\bullet\bullet}$$

$$\hat{\alpha}_i = \bar{Y}_{i\bullet\bullet} - \hat{\mu} = \bar{Y}_{i\bullet\bullet} - \bar{Y}_{\bullet\bullet\bullet}$$

$$\hat{\beta}_j = \bar{Y}_{\bullet j\bullet} - \hat{\mu} = \bar{Y}_{\bullet j\bullet} - \bar{Y}_{\bullet\bullet\bullet}$$

$$\hat{\gamma}_{ij} = \widehat{(\alpha\beta)}_{ij} = \bar{Y}_{ij\bullet} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j = \bar{Y}_{ij\bullet} - \bar{Y}_{\bullet\bullet\bullet} - (\bar{Y}_{i\bullet\bullet} - \bar{Y}_{\bullet\bullet\bullet}) - (\bar{Y}_{\bullet j\bullet} - \bar{Y}_{\bullet\bullet\bullet}) = \bar{Y}_{ij\bullet} - \bar{Y}_{i\bullet\bullet} - \bar{Y}_{\bullet j\bullet} + \bar{Y}_{\bullet\bullet\bullet}$$

where

$\bar{Y}_{\bullet\bullet\bullet} = \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K Y_{ijk}/(IJK)$ is the grand mean of all response variables.

$\bar{Y}_{i\bullet\bullet} = \sum_{j=1}^J \sum_{k=1}^K Y_{ijk}/(JK)$ denotes the mean of the i th level of factor A, combining all levels of factor B.

$\bar{Y}_{\bullet j\bullet} = \sum_{i=1}^I \sum_{k=1}^K Y_{ijk}/(IK)$ denotes the mean of the j th level of factor B, combining all levels of factor A.

$\bar{Y}_{ij\bullet} = \sum_{k=1}^K Y_{ijk}/K$ denotes the mean of the group under i th level of factor A and j th level of factor B.

- The **additive model**, *a.k.a.* the **main effects model** (without interaction) has simpler parametrization,

$$\mu_{ij} = \mu + \alpha_i + \beta_j$$

- As in one-way ANOVA, there are various ways to set constraints to eliminate parameter redundancy.

For example, the constraints

$$\alpha_1 = 0, \quad \beta_1 = 0, \quad \gamma_{1j} = 0, \quad \gamma_{i1} = 0$$

are commonly used in software such as R.

- Model parameters have different meaning under different parameter constraints.
However the estimated treatment effects and result interpretations stay the same.
- The linear model equation represents a set of equations, one for each observation.

Example

Consider a two-factor study with balanced design, where factor A has $I = 2$ levels, factor B has $J = 3$ levels, and common replicates $K_{ij} = K = 2$. The linear model

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \sim i.i.d. \ N(0, \sigma^2)$$

represents 12 observation level equations.

$$\left\{ \begin{array}{l} y_{111} = \mu + \alpha_1 + \beta_1 + \gamma_{11} + \varepsilon_{111} \\ y_{112} = \mu + \alpha_1 + \beta_1 + \gamma_{11} + \varepsilon_{112} \\ y_{121} = \mu + \alpha_1 + \beta_2 + \gamma_{12} + \varepsilon_{121} \\ y_{122} = \mu + \alpha_1 + \beta_2 + \gamma_{12} + \varepsilon_{122} \\ y_{131} = \mu + \alpha_1 + \beta_3 + \gamma_{13} + \varepsilon_{131} \\ y_{132} = \mu + \alpha_1 + \beta_3 + \gamma_{13} + \varepsilon_{132} \\ y_{211} = \mu + \alpha_2 + \beta_1 + \gamma_{21} + \varepsilon_{211} \\ y_{212} = \mu + \alpha_2 + \beta_1 + \gamma_{21} + \varepsilon_{212} \\ y_{221} = \mu + \alpha_2 + \beta_2 + \gamma_{22} + \varepsilon_{221} \\ y_{222} = \mu + \alpha_2 + \beta_2 + \gamma_{22} + \varepsilon_{222} \\ y_{231} = \mu + \alpha_2 + \beta_3 + \gamma_{23} + \varepsilon_{231} \\ y_{232} = \mu + \alpha_2 + \beta_3 + \gamma_{23} + \varepsilon_{232} \end{array} \right.$$

There are $I \times J = 6$ treatment combinations, but we have assigned 12 parameters:

$$\begin{array}{ll} \text{the grand mean:} & \mu \\ \text{A main effects:} & \alpha_1, \alpha_2 \\ \text{B main effects:} & \beta_1, \beta_2, \beta_3 \\ \text{A*B interactions:} & \gamma_{11}, \gamma_{12}, \gamma_{13}, \gamma_{21}, \gamma_{22}, \gamma_{23} \end{array}$$

There is obvious redundancy. Over parametrization makes the parameters unidentifiable.

Without imposing further constraints, there would be infinitely many solutions of the parameters.

Consider using the zero-sum constraints on the main effect coefficients

$$\sum_{i=1}^I \alpha_i = 0, \quad \sum_{j=1}^J \beta_j = 0$$

and on the interaction terms,

$$\sum_{i=1}^I \gamma_{ij} = 0, \quad \forall j = 1, 2, 3; \quad \sum_{j=1}^J \gamma_{ij} = 0, \quad \forall i = 1, 2.$$

The constraints imply

$$\begin{array}{ll} \alpha_2 = -\alpha_1 & (\text{from } \sum_i \alpha_i = 0) \\ \beta_3 = -\beta_1 - \beta_2 & (\text{from } \sum_i \beta_i = 0) \\ \gamma_{21} = -\gamma_{11} & (\text{from } \sum_i \gamma_{i1} = 0) \\ \gamma_{22} = -\gamma_{12} & (\text{from } \sum_i \gamma_{i2} = 0) \\ \gamma_{13} = -\gamma_{11} - \gamma_{12} & (\text{from } \sum_j \gamma_{1j} = 0) \\ \gamma_{23} = -\gamma_{21} - \gamma_{22} = \gamma_{11} + \gamma_{12} & (\text{from } \sum_j \gamma_{2j} = 0) \end{array}$$

The number of (independent) parameters is appropriately reduced to $6 = I \times J$: $\mu, \alpha_1, \beta_1, \beta_2, \gamma_{11}, \gamma_{12}$.
(Note that the redundant constraint $\sum_i \gamma_{i3} = 0$ confirms $\gamma_{13} = -\gamma_{23}$, adding no new information.)

Under the set of constraints, the linear model

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \varepsilon_{ijk}$$

now has the form

$$\left\{ \begin{array}{l} y_{111} = \mu + \alpha_1 + \beta_1 + \gamma_{11} + \varepsilon_{111} \\ y_{112} = \mu + \alpha_1 + \beta_1 + \gamma_{11} + \varepsilon_{112} \\ y_{121} = \mu + \alpha_1 + \beta_2 + \gamma_{12} + \varepsilon_{121} \\ y_{122} = \mu + \alpha_1 + \beta_2 + \gamma_{12} + \varepsilon_{122} \\ y_{131} = \mu + \alpha_1 - \beta_1 - \beta_2 - \gamma_{11} - \gamma_{12} + \varepsilon_{131} \\ y_{132} = \mu + \alpha_1 - \beta_1 - \beta_2 - \gamma_{11} - \gamma_{12} + \varepsilon_{132} \\ y_{211} = \mu - \alpha_1 + \beta_1 - \gamma_{11} + \varepsilon_{211} \\ y_{212} = \mu - \alpha_1 + \beta_1 - \gamma_{11} + \varepsilon_{212} \\ y_{221} = \mu - \alpha_1 + \beta_2 - \gamma_{12} + \varepsilon_{221} \\ y_{222} = \mu - \alpha_1 + \beta_2 - \gamma_{12} + \varepsilon_{222} \\ y_{231} = \mu - \alpha_1 - \beta_1 - \beta_2 + \gamma_{11} + \gamma_{12} + \varepsilon_{231} \\ y_{232} = \mu - \alpha_1 - \beta_1 - \beta_2 + \gamma_{11} + \gamma_{12} + \varepsilon_{232} \end{array} \right.$$

In vector-matrix form with design matrix denoted as $\mathbf{X}$, the model can be expressed as

$$Y = \mathbf{X}\beta + \varepsilon$$

which represents

$$\begin{bmatrix} y_{111} \\ y_{112} \\ y_{121} \\ y_{122} \\ y_{131} \\ y_{132} \\ y_{211} \\ y_{212} \\ y_{221} \\ y_{222} \\ y_{231} \\ y_{232} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & -1 & 1 & 0 & -1 & 0 \\ 1 & -1 & 1 & 0 & -1 & 0 \\ 1 & -1 & 0 & 1 & 0 & -1 \\ 1 & -1 & 0 & 1 & 0 & -1 \\ 1 & -1 & -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \mu \\ \alpha_1 \\ \beta_1 \\ \beta_2 \\ \gamma_{11} \\ \gamma_{12} \end{bmatrix} + \begin{bmatrix} \varepsilon_{111} \\ \varepsilon_{112} \\ \varepsilon_{121} \\ \varepsilon_{122} \\ \varepsilon_{131} \\ \varepsilon_{132} \\ \varepsilon_{211} \\ \varepsilon_{212} \\ \varepsilon_{221} \\ \varepsilon_{222} \\ \varepsilon_{231} \\ \varepsilon_{232} \end{bmatrix}$$

The horizontal lines indicate the replicates within each treatment combination.

Sums of squares for two-way layout

For the main effects and interaction model

$$y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \sim i.i.d. N(0, \sigma^2)$$

with parameter constraints

$$\sum_i \alpha_i = 0, \quad \sum_j \beta_j = 0, \quad \sum_i \gamma_{ij} = 0, \quad \sum_j \gamma_{ij} = 0,$$

when the design is **balanced**, that is, when early treatment combination has the same number of replicates with

$$K_{ij} \equiv K,$$

the main effects for A and B are orthogonal to each other and to $A * B$. In the balanced design matrix, the block of columns with respect to A , the block of columns w.r.t. B , and the block w.r.t. $A * B$ are mutually independent.

The orthogonality in balanced design allows us to determine a sum of squares for factors A, B and their interaction $A * B$:

$$\begin{aligned} SS_A &= \sum_i \sum_j \sum_k (\bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet\bullet\bullet})^2 = KJ \sum_i \left(\underbrace{\bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet\bullet\bullet}}_{\hat{\alpha}_i} \right)^2 \\ SS_B &= \sum_i \sum_j \sum_k (\bar{y}_{\bullet j \bullet} - \bar{y}_{\bullet\bullet\bullet})^2 = KI \sum_j \left(\underbrace{\bar{y}_{\bullet j \bullet} - \bar{y}_{\bullet\bullet\bullet}}_{\hat{\beta}_j} \right)^2 \\ SS_{A*B} &= \sum_i \sum_j \sum_k (\bar{y}_{ij\bullet} - \bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet j \bullet} + \bar{y}_{\bullet\bullet\bullet})^2 \\ &= \sum_{i,j,k} \left(\underbrace{\bar{y}_{ij\bullet} - \left[\underbrace{\bar{y}_{\bullet\bullet\bullet}}_{\hat{\mu}} + \underbrace{(\bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet\bullet\bullet})}_{\hat{\alpha}_i} + \underbrace{(\bar{y}_{\bullet j \bullet} - \bar{y}_{\bullet\bullet\bullet})}_{\hat{\beta}_j} \right]}_{\hat{\gamma}_{ij}} \right)^2 \end{aligned}$$

In the above, the indicated parameter estimates $(\hat{\mu}, \hat{\alpha}_i, \hat{\beta}_j, \hat{\gamma}_{ij})$ are subjected to the zero-sum constraints in (1).

Two-way ANOVA table

For balanced, two-factorial data, the results of analysis of variance are commonly displayed in an ANOVA table.

For the main effect-interaction model

$$y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \varepsilon_{ijk}, \quad i = 1, \dots, I; \quad j = 1, \dots, J$$

with replication $K_{ij} = K$, the two-way ANOVA table has the following form.

Source	d.f.	SS	MS	Variance ratio (F)
Model	$IJ - 1$	SS_{Model}	$SS_{Model}/(IJ - 1)$	MS_{Model}/MS_E
A (I-levels)	$I - 1$	SS_A	$SS_A/(I - 1)$	MS_A/MS_E
B (J-levels)	$J - 1$	SS_B	$SS_B/(J - 1)$	MS_B/MS_E
A*B	$(I - 1)(J - 1)$	SS_{AB}	$SS_{AB}/[(I - 1)(J - 1)]$	MS_{AB}/MS_E
Error	$(K - 1)IJ$	SS_E	$SS_E/[(K - 1)IJ]$	
Total	$IJK - 1$	SS_{Total}		

Remarks (on the two-way table with interaction)

- In the "Model" row, $SS_{Model} = SS_A + SS_B + SS_{AB}$.
- The "Model" row may not show up in some software output.

Expected value of the sums of squares

Analogous to the derivations for one-way layout,

$$\begin{aligned} \mathbb{E}(MS_E) &= \sigma^2 \\ \mathbb{E}(MS_A) &= \sigma^2 + \frac{JK}{I-1} \sum_{i=1}^I \alpha_i^2 \\ \mathbb{E}(MS_B) &= \sigma^2 + \frac{IK}{J-1} \sum_{j=1}^J \beta_j^2 \\ \mathbb{E}(MS_{AB}) &= \sigma^2 + \frac{K}{(I-1)(J-1)} \sum_{i=1}^I \sum_{j=1}^J \gamma_{ij}^2 \end{aligned}$$

Two-way hypothesis testing

Assuming the observations Y_{ij} are $i.i.d. \sim N(\mu_i, \sigma^2)$.

- To test $H_A : \alpha_i \equiv 0, i = 1, \dots, I$, use test statistic $F = \frac{MS_A}{MS_E}$, which $\sim F_{df_A, df_E}$ under the null H_A .

- To test $H_B : \beta_i \equiv 0, j = 1, \dots, J$, use test statistic $F = \frac{MS_B}{MS_E}$, which $\sim F_{df_B, df_E}$ under the null H_B .
- To test $H_{AB} : \gamma_{ij} \equiv 0, \forall i, j$, use test statistic $F = \frac{MS_{AB}}{MS_E}$, which $\sim F_{df_{AB}, df_E}$ under the null H_{AB} .

Theoretical results (Basic Theorem and extensions for sums of squares in analysis of variance)

Under each of the null hypothesis H_* ,

- All MS_* 's, that is, the MS_* in the numerator and the MS_* in the denominator, are mutually independent.
- Each $MS_*/\sigma^2 \sim \chi_{df_*}^2$ under the corresponding null hypothesis.
- $MS_E/\sigma^2 \sim \chi_{df_E}^2$.

The relation between Two-way and One-way ANOVA

Two-way analysis looks into the factor structure of the data.

Recall the analysis of variance table for one-factorial analysis:

Source	d.f.	SS	MS	Variance ratio (F)
Treatment(g-levels)	$g - 1$	SS_{Trt}	$SS_{Trt}/(g - 1)$	MS_{Trt}/MS_E
Error	$n - g$	SS_E	$SS_E/(n - g)$	
total	$n - 1$	SS_{Total}		

where g is the number of levels of the treatment, or number of treatments.

Two-way data analysis of variance is related to one-way ANOVA in several ways.

- If we do not consider the factor structure, regarding the data as one treatment with $g = I \times J$ levels, the analysis of variance is reduced to one-way. For balanced data $K_{ij} \equiv K, n = IJK = gK$, the ANOVA table becomes

Source	d.f.	SS	MS	Variance ratio (F)
Treatment	$IJ - 1$	SS_{Trt}	$SS_{Trt}/(IJ - 1)$	MS_{Trt}/MS_E
Error	$(K - 1)IJ$	SS_E	$SS_E/[(K - 1)IJ]$	
total	$IJK - 1$	SS_{Total}		

Note that we have the same degree of freedom to estimate the error. No gain on that regard.

- we may come across the situation that one of the two factors is not significant in explaining the variations in the observations.

For example, if we do not consider the factor B, regarding the data as under treatment A only, with $g = I$ treatment levels, the analysis of variance is reduced to one-way with factor A only.

For balanced data $K_{ij} \equiv K, n = IJK = gJK$, the one-way ANOVA table is

Source	d.f.	SS	MS	F
Treatment A (I-levels)	$I - 1$	SS_A	$SS_A/(I - 1)$	MS_A/MS_E
Error	$IJK - I$	SS_E	$SS_E/(IJK - I)$	
total	$IJK - 1$	SS_{Total}		

Note that we have larger degree of freedom to estimate the error, since different B-levels are treated as replicates within each level of A.

Two-way ANOVA table without interaction

For balanced two-factor measurements with replicates $K_{ij} = K$, when the interaction between factor A and factor B is not significant in explaining the remaining variation in data after subtracting the effect of factors A and B, the two-way ANOVA model without interaction term is

$$y_{ijk} = \mu + \alpha_i + \beta_j + \varepsilon_{ijk}, \quad i = 1, \dots, I; j = 1, \dots, J$$

This is called the main effects model.

The two-way ANOVA table for the main effects model has the form

Source	d.f.	SS	MS	Variance ratio (F)
A (I-levels)	$I - 1$	SS_A	$SS_A/(I - 1)$	MS_A/MS_E
B (J-levels)	$J - 1$	SS_B	$SS_B/(J - 1)$	MS_B/MS_E
Error	$IJK - I - J + 1$	SS_E	$SS_E/(IJK - I - J + 1)$	
Total	$IJK - 1$	SS_{Total}		

Remarks (on the two-way table with $\gamma_{ij} = 0$)

Compared with the two-way ANOVA table assuming interaction $\gamma_{ij} \neq 0$,

- The SS_E for the main effects model with $\gamma_{ij} = 0$ is the sum $SS_{AB} + SS_E$ in the table with interaction.
- The degree of freedom of the error df_E for the model with $\gamma_{ij} = 0$ is also $df_{AB} + df_E$ in the ANOVA table with interaction.

Two-factor layout without replication

Due to resource constraints, many two-way experiments only have one measurement obtained for each treatment combination. A two-way analysis of variance can be carried out on two factor data with no replication, with limitations.

For data with no replication, $K_{ij} \equiv 1$. The third subscript k is no longer needed and can be dropped. The data structure simplifies to

	B-level 1	B-level 2		B-level b
A-level 1	y_{11}	y_{12}		y_{1b}
A-level 2	y_{21}	y_{22}		y_{2b}
$\vdots$	$\vdots$	$\vdots$		$\vdots$
A-level a	y_{a1}	y_{a2}		y_{ab}

The two factor linear model for no-replication case has two subscripts instead of three,

$$Y_{ij} = \mu_{ij} + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim i.i.d. N(0, \sigma^2), \quad i = 1, \dots, I; j = 1, \dots, J.$$

(Note: Without replicates, further assumptions are needed to identify or separate the two terms on the right side of the model formula.)

Example — No replication two-way layout ANOVA

The 12 data points in the table are measurements y_{ij} from an experiment with two factors of levels $I = 3, J = 4$ respectively. There is no replication. Each treatment combination only has one measurement.

y_{ij}	j=1	j=2	j=3	j=4	row-sum
i=1	3	2	1	2	(8)
i=2	8	7	8	5	(28)
i=3	7	6	9	2	(24)
col-sum	(18)	(15)	(18)	(9)	60

The added marginal sum column and row are for the convenience of parameter estimation below.

A general two-way linear model with no replication is

$$Y_{ij} = \mu_{ij} + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim i.i.d. N(0, \sigma^2), \quad i = 1, 2, 3; j = 1, 2, 3, 4.$$

Since there are no replicates, the cell level mean estimates are the observed values:

$$\hat{\mu}_{ij} = y_{ij}$$

Suppose we impose the structure

$$\mu_{ij} = \mu + \alpha_i + \beta_j$$

That is, we consider fitting the additive model

$$Y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij} \quad \varepsilon_{ij} \sim i.i.d. N(0, \sigma^2), \quad i = 1, 2, 3; j = 1, 2, 3, 4.$$

We choose to use the constraints

$$\sum_{i=1}^3 \alpha_i = 0, \quad \sum_{j=1}^4 \beta_j = 0.$$

The grand mean μ can be estimated by the LME

$$\hat{\mu} = \bar{y}_{\bullet\bullet\bullet} = \bar{y}_{\bullet\bullet} = \frac{1}{12} \sum_{i=1}^3 \sum_{j=1}^4 y_{ij} = \frac{60}{12} = 5$$

Parameters α_i and β_j , the row effects and column (differential) effects, can be estimated by

$$\begin{cases} \hat{\alpha}_i &= \bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet\bullet\bullet} = \bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet} \\ \hat{\beta}_j &= \bar{y}_{\bullet j\bullet} - \bar{y}_{\bullet\bullet\bullet} = \bar{y}_{\bullet j} - \bar{y}_{\bullet\bullet} \end{cases}$$

To obtain and view the estimates using the two-way layout table, lest's first create the table with entries

$$\hat{\mu}_{ij} - \hat{\mu} = y_{ij} - \bar{y}_{\bullet\bullet}$$

which can be obtained from the original data table with the grand mean subtracted from each entry.

$\hat{\mu}_{ij} - \hat{\mu}$	j=1	j=2	j=3	j=4	row-sum	$\hat{\alpha}_i$
i=1	-2	-3	-4	-3	(-12)	-3
i=2	3	2	3	0	(8)	2
i=3	2	1	4	-3	(4)	1
col-sum	(3)	(0)	(3)	(-6)		0
$\hat{\beta}_j$	1	0	1	-2		

Parameter estimates $\hat{\alpha}_i$ and $\hat{\beta}_j$ are just row and column averages in the above table with entry $\hat{\mu}_{ij} - \hat{\mu} = y_{ij} - \bar{y}_{\bullet\bullet}$.

Further subtracting the main effect estimates $\hat{\alpha}_i + \hat{\beta}_j$, we are left with

$\hat{\mu}_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j$	j=1	j=2	j=3	j=4
i=1	0	0	-2	2
i=2	0	0	0	0
i=3	0	0	2	-2

The table shows the estimated residuals

$$\hat{\varepsilon}_{ij} = \hat{\mu}_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j$$

Suppose that we were ambitious and would like to fit the main effects plus interaction model

$$Y_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim i.i.d. N(0, \sigma^2)$$

Then the estimated interaction term would be

$$\hat{\gamma}_{ij} = \hat{\mu}_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j$$

The estimated residuals

$$\hat{\varepsilon}_{ij} = y_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j - \hat{\gamma}_{ij} = \hat{\mu}_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j - \hat{\gamma}_{ij}$$

resulting in

$$\hat{\varepsilon}_{ij} = 0, \quad \forall i, j.$$

This does not imply that there are no noise in the data. It only indicates that by fitting the model with interaction term, there are no degrees of freedom left to estimate the variance σ^2 , that is, we are not capable to estimate the noise variation.

The model with interaction has a “perfect” but dubious, flawed fit, since we can't estimate the variance σ^2 .

For two-way data with no replication, it is suitable to use the main effects model

$$y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}, \quad i = 1, \dots, I; j = 1, \dots, J$$

The two-way ANOVA table with no replication has the form

Source	d.f.	SS	MS	Variance ratio (F)
A (I-levels)	$I - 1$	SS_A	$SS_A/(I - 1)$	MS_A/MS_E
B (J-levels)	$J - 1$	SS_B	$SS_B/(J - 1)$	MS_B/MS_E
Error	$(I - 1)(J - 1)$	SS_E	$SS_E/[(I - 1)(J - 1)]$	
Total	$IJ - 1$	SS_{Total}		

Related reference: Rice 12.3; Casella and Berger chapter 11.